Financial Management

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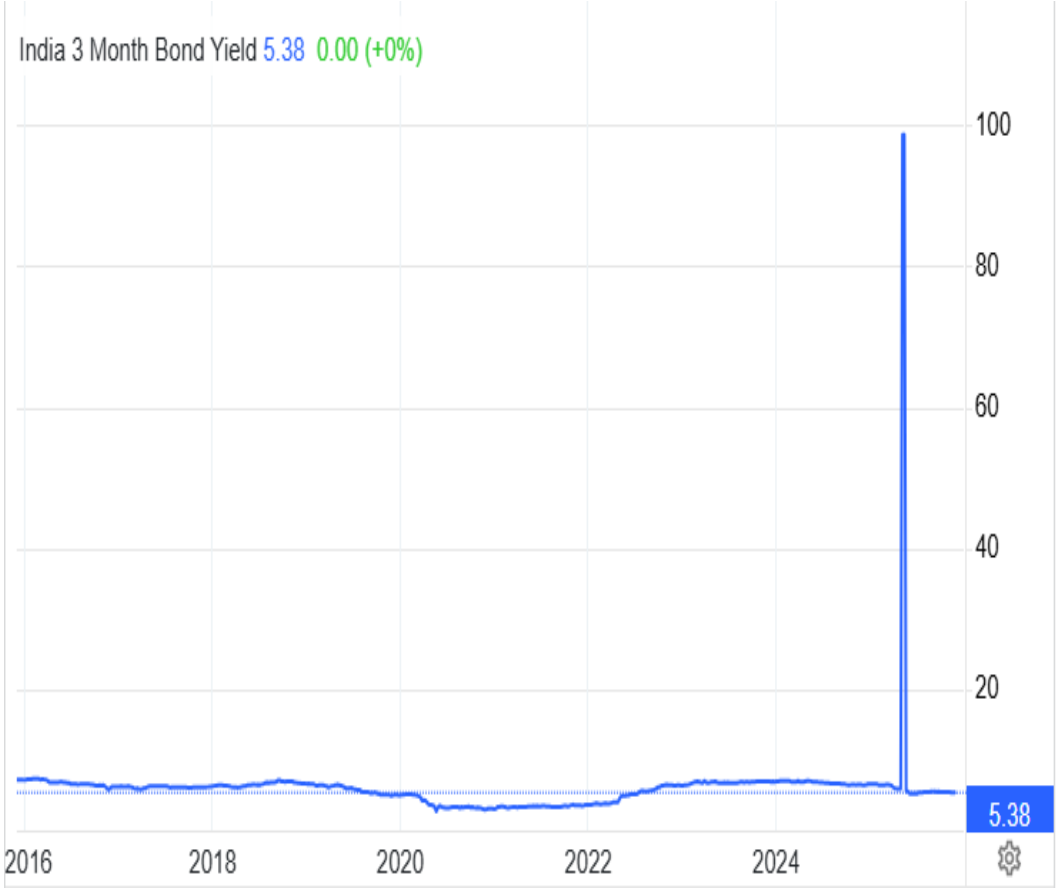
Module 2: Part 3
Term Structure of Interest Rates

Topics

1. Interest Rates Fundamentals
2. Loanable Fund Theory
3. Determinants of Interest Rates for Individual Securities
4. Term Structure of Interest Rates : Theories
5. Implications of the Term Structure of Interest Rates

Importance of Interest Rate

- Economic decision of the household
- Economic decision of firms and businesses
- Overall performance of the economy
- Eg: India 3 Month Bond Yield

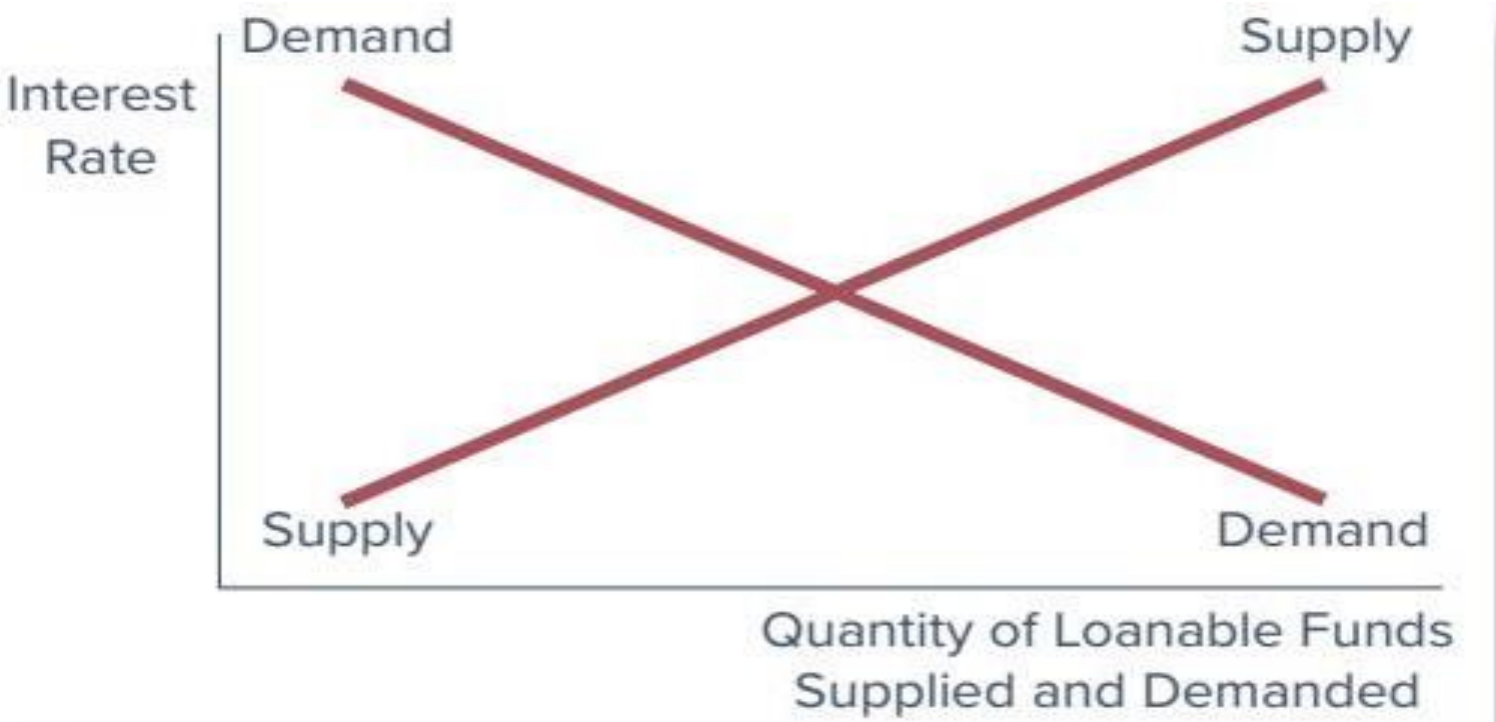


Source: Tradingeconomics.com

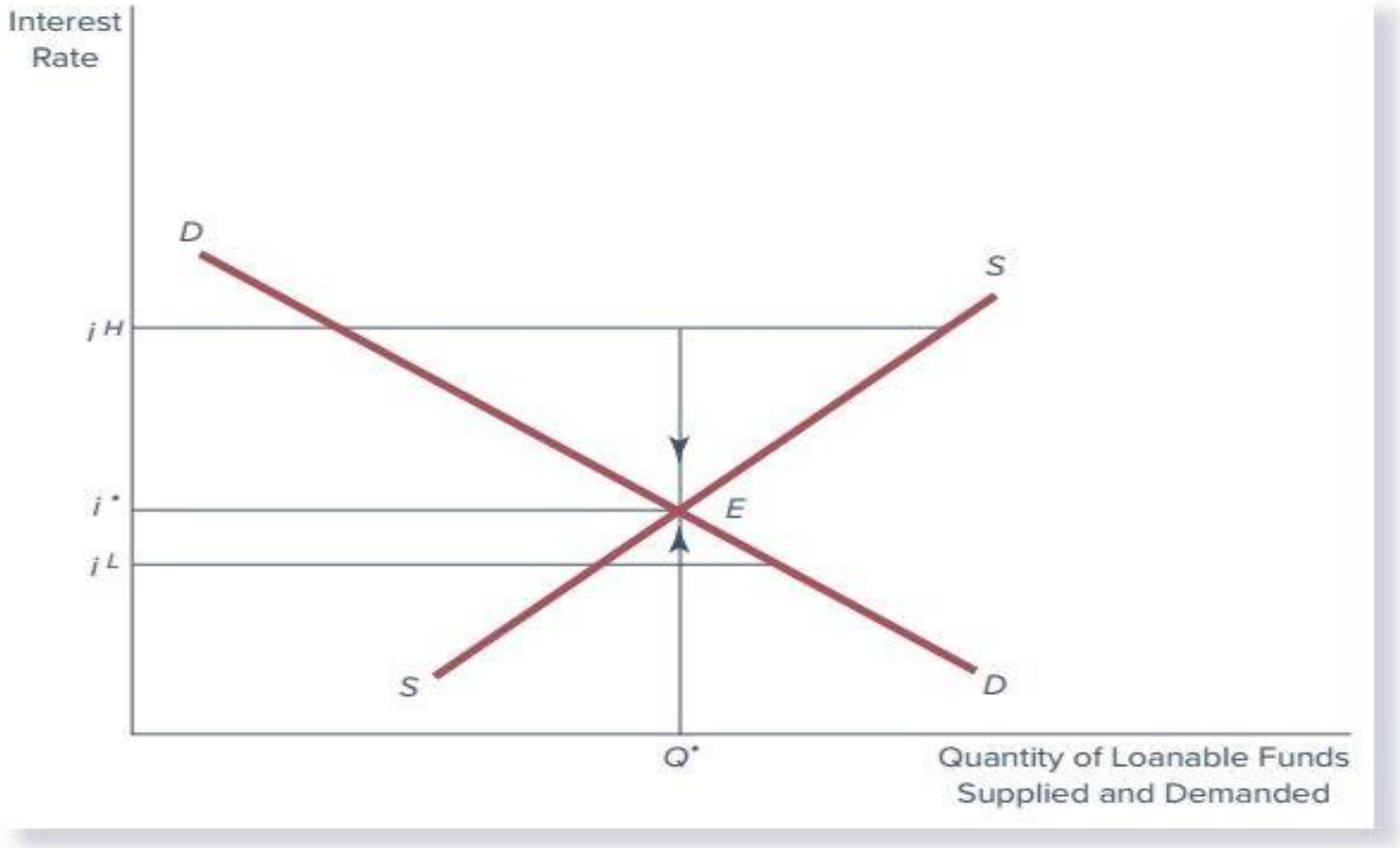
Interest Rates & Its Movements: Loanable Fund Theory

- Loanable funds theory explains interest rates and interest rate movements
- Views level of interest rates in financial markets as a result of the supply and demand for loanable funds
- Domestic and foreign households, businesses, and governments all supply and demand loanable funds

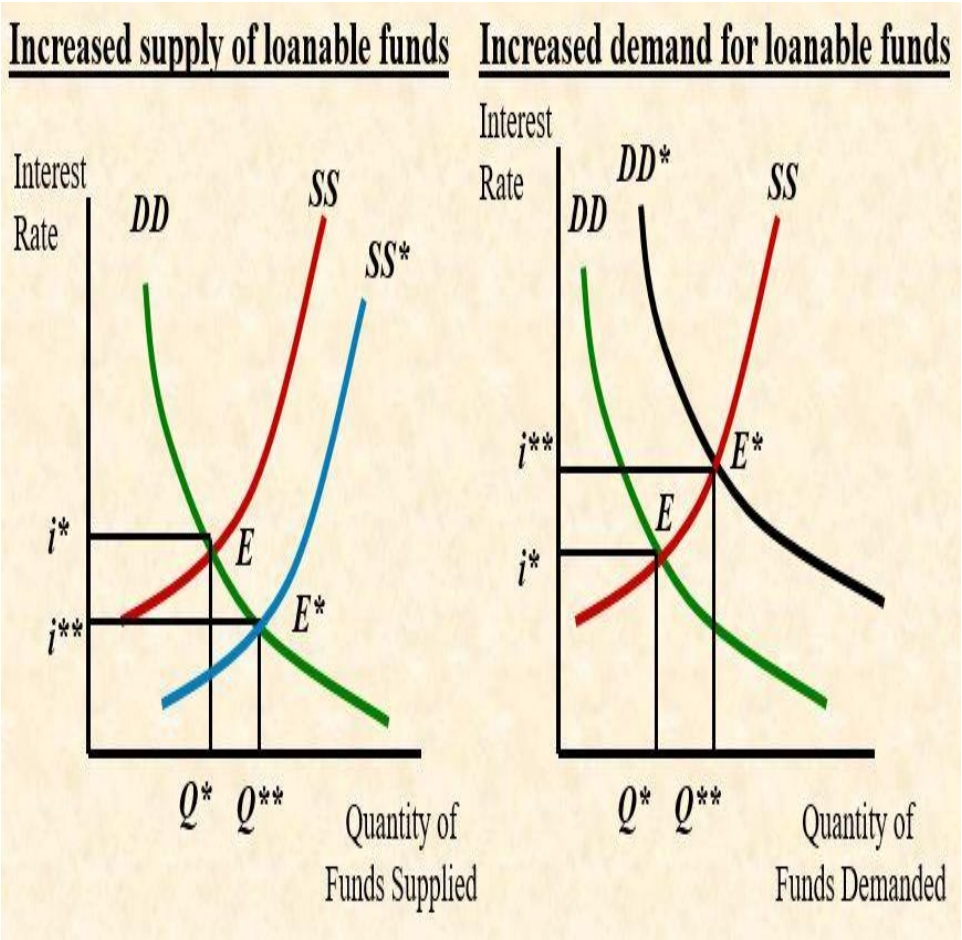
Supply and Demand for Loanable Fund



Loanable Fund Theory: Determining Equilibrium Interest Rates



Factors causing shift in Supply and Demand for Loanable Funds



Panel A: The supply of funds

Factor	Impact on Supply of Funds	Impact on Equilibrium Interest Rate*
Interest rate (+)	Movement along the supply curve	Direct
Total wealth (+)	Shift supply curve	Inverse
Risk of financial security (−)	Shift supply curve	Direct
Near-term spending needs (−)	Shift supply curve	Direct
Monetary expansion (+)	Shift supply curve	Inverse
Economic conditions (+)	Shift supply curve	Inverse

Panel B: The demand for funds

Factor	Impact on Demand for Funds	Impact on Equilibrium Interest Rate
Interest rate (−)	Movement along the demand curve	Direct
Utility derived from asset purchased with borrowed funds (+)	Shift demand curve	Direct
Restrictiveness of nonprice conditions (−)	Shift demand curve	Inverse
Economic conditions (+)	Shift demand curve	Direct

*A "direct" impact on equilibrium interest rates means that as the "factor" increases (decreases) the equilibrium interest rate increases (decreases). An "inverse" impact means that as the factor increases (decreases) the equilibrium interest rate decreases (increases).

Determinants of Interest Rates for Individual Securities

- Inflation
- Real risk-free rate
- Default Risk
- Liquidity Risk
- Special Provisions (Taxability, Convertibility, and Callability)
- Term to Maturity

Determinants of Interest Rates for Individual Securities

Inflation—the continual increase in the price level of a basket of goods and services.

Real risk-free rate—nominal risk-free rate that would exist on a security if no inflation were expected.

Default risk—risk that a security issuer will default on the security by missing an interest or principal payment.

Liquidity risk—risk that a security cannot be sold at a predictable price with low transaction costs at short notice.

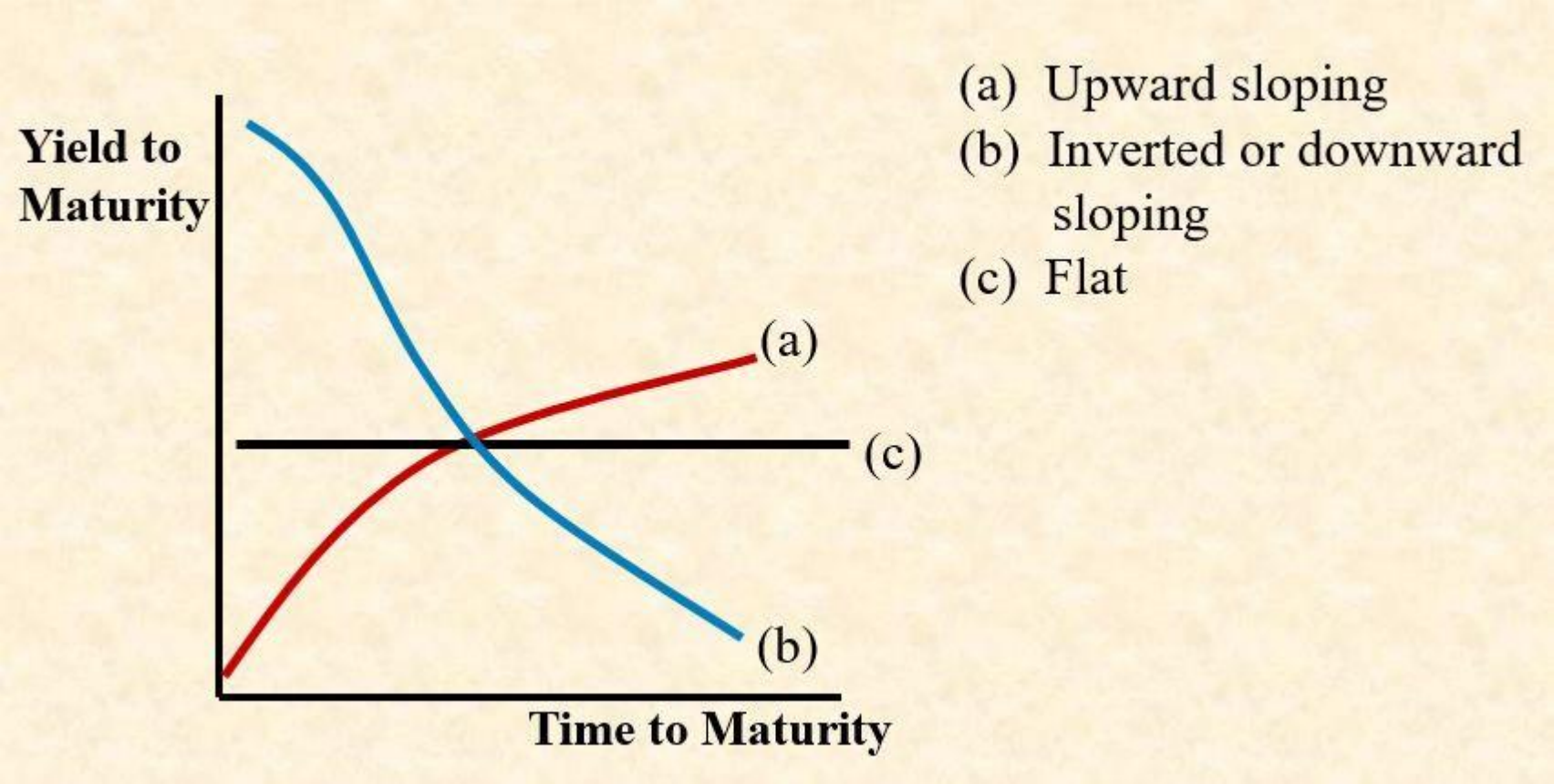
Special provisions—provisions (e.g., taxability, convertibility, and callability) that impact the security holder beneficially or adversely and as such are reflected in the interest rates on securities that contain such provisions.

Term to maturity—length of time a security has until maturity.

Term Structure of Interest Rates: The Yield Curve

- Securities with identical risk, liquidity, and tax characteristics may have different interest rates because the time remaining to maturity is different.
- Yield curve: a plot of the yield on bonds with differing terms to maturity but the same risk, liquidity and tax considerations
  - Upward-sloping: long-term rates are above short-term rates
  - Flat: short- and long-term rates are the same
  - Inverted: long-term rates are below short-term rates

Term Structure of Interest Rates: The Yield Curve



Types of Yield Curves and its Implications

Normal

- long term yields are higher than short term yields
- signals that the economy is in an expansionary mode

Flat

- very little variation between short and long term yields
- Signals that the market is unsure about the future direction of the economy

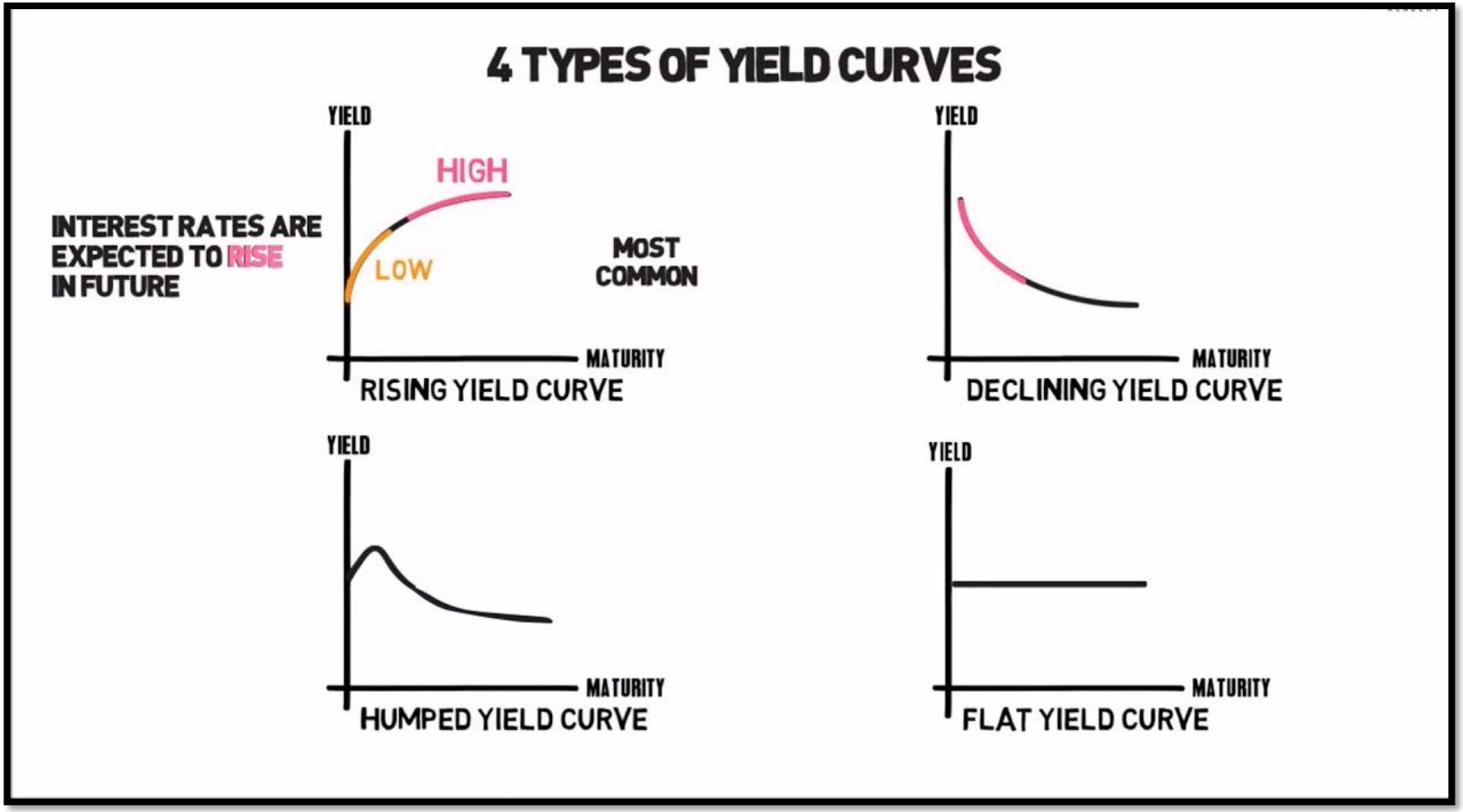
Inverted

- short term yields are higher than long term yields

Humped

- interest rates on medium-term fixed income securities are higher than the rates of both long and short-term instruments

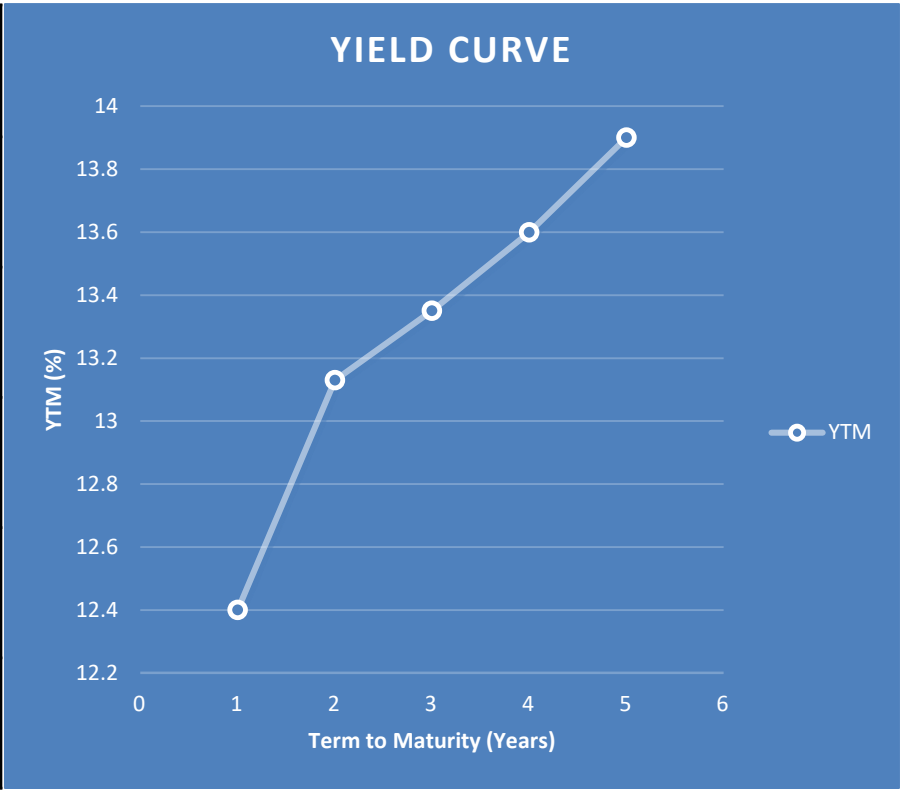
Types of Yield Curve



Yield Curve

Consider the following data of the government securities

Face Value	Interest Rate	Current Price	Maturity (yrs)	YTM
100000	0	88968	1	12.4
100000	12.75	99367	2	13.13
100000	13.5	100352	3	13.35
100000	13.5	99706	4	13.6
100000	13.75	99484	5	13.9



Theories of Term Structure of Interest Rates

- The theory of the term structure of interest rates must explain the following facts:
  1. Interest rates on bonds of different maturities move together over time.
  2. When short-term interest rates are low, yield curves are more likely to have an upward slope; when short-term rates are high, yield curves are more likely to slope downward and be inverted.
  3. Yield curves almost always slope upward.

Term Structure of Interest Rate: Theories

Three theories to explain the three facts:

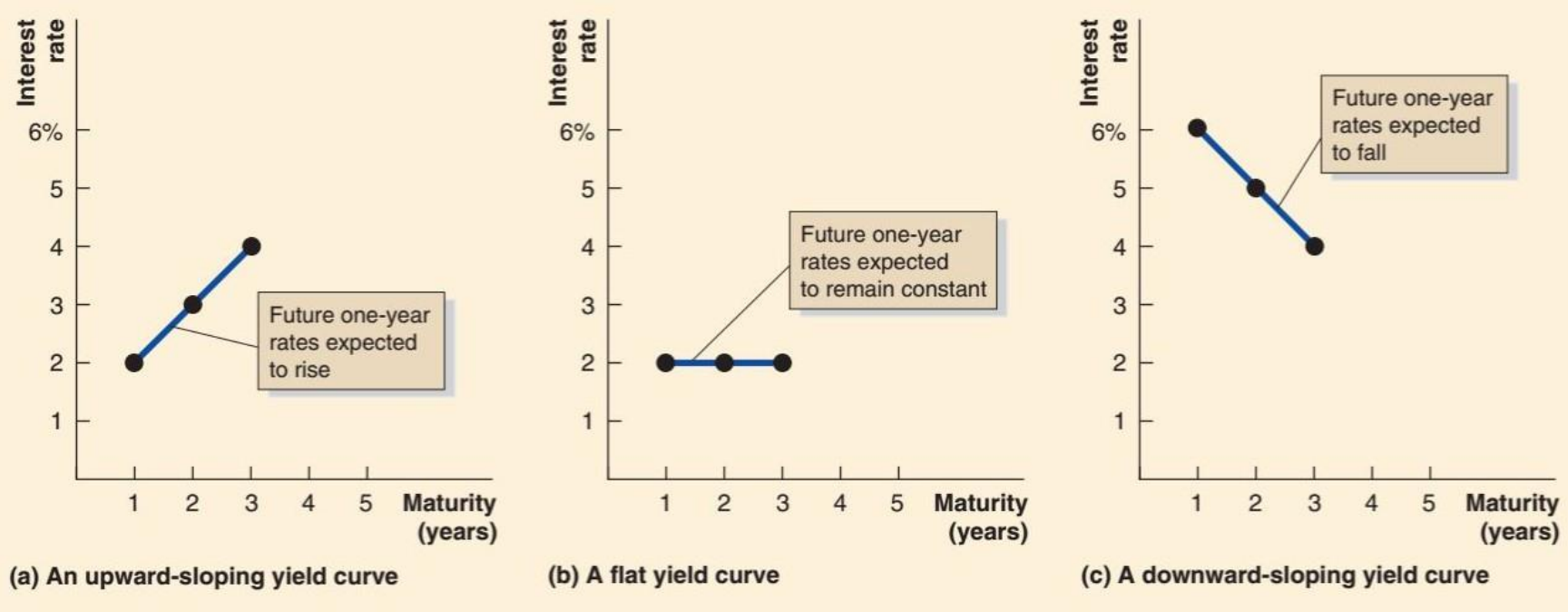
1. Expectations theory explains the first two facts but not the third.
2. Liquidity premium theory combines the two theories to explain all three facts.
3. Segmented markets theory explains the third fact but not the first two.

Theories of Term Structure of Interest Rates

1. Expectations Theory:

- ***Assumes that investors know with certainty what lies ahead of them.***
- Any long-term rate is equal o the geometric mean of current and future one-year rates expected by the market participants.
- The shape of the yield curve can be explained by the interest rate expectations of those who participate in the market.
  - If the yield curve is upward sloping, short-term rates are expected to rise.
  - If the curve is downward sloping, short-term rates are expected to fall.
  - A flat yield curve implies that the market expects short-term rates to remain constant

Using Yield Curve to Explain Expectations Theory



Limitations of Expectation Theory

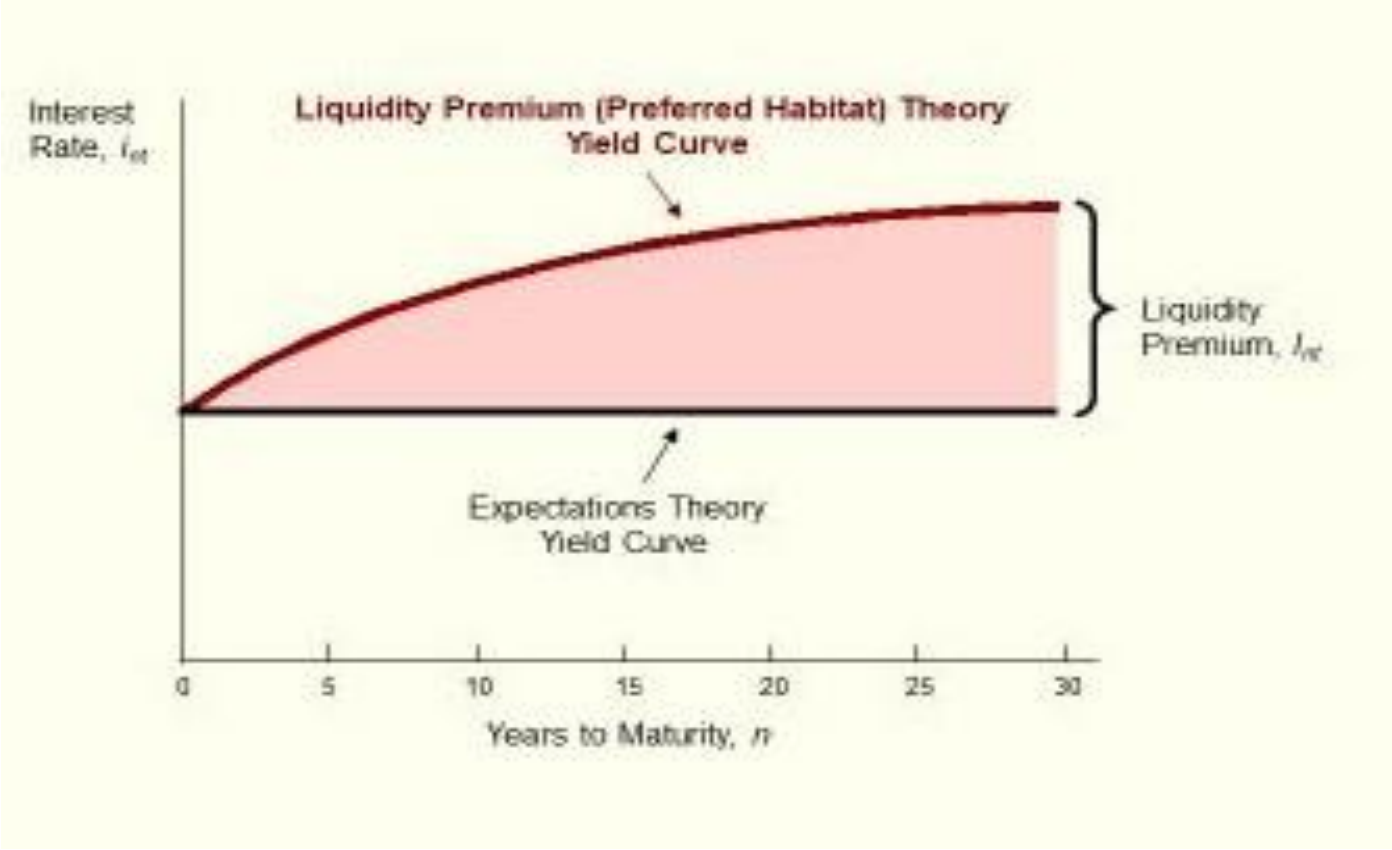
- Expectations theory explains:
  - Why the term structure of interest rates changes at different times.
  - Why interest rates on bonds with different maturities move together over time (fact 1).
  - Why yield curves tend to slope up when
    - short-term rates are low and slope down when short-term rates are high (fact 2).
- Cannot explain why yield curves usually slope upward (fact 3)

Theories of Term Structure of Interest Rates

2. Liquidity Preference Theory (Preferred Habitat Theory):

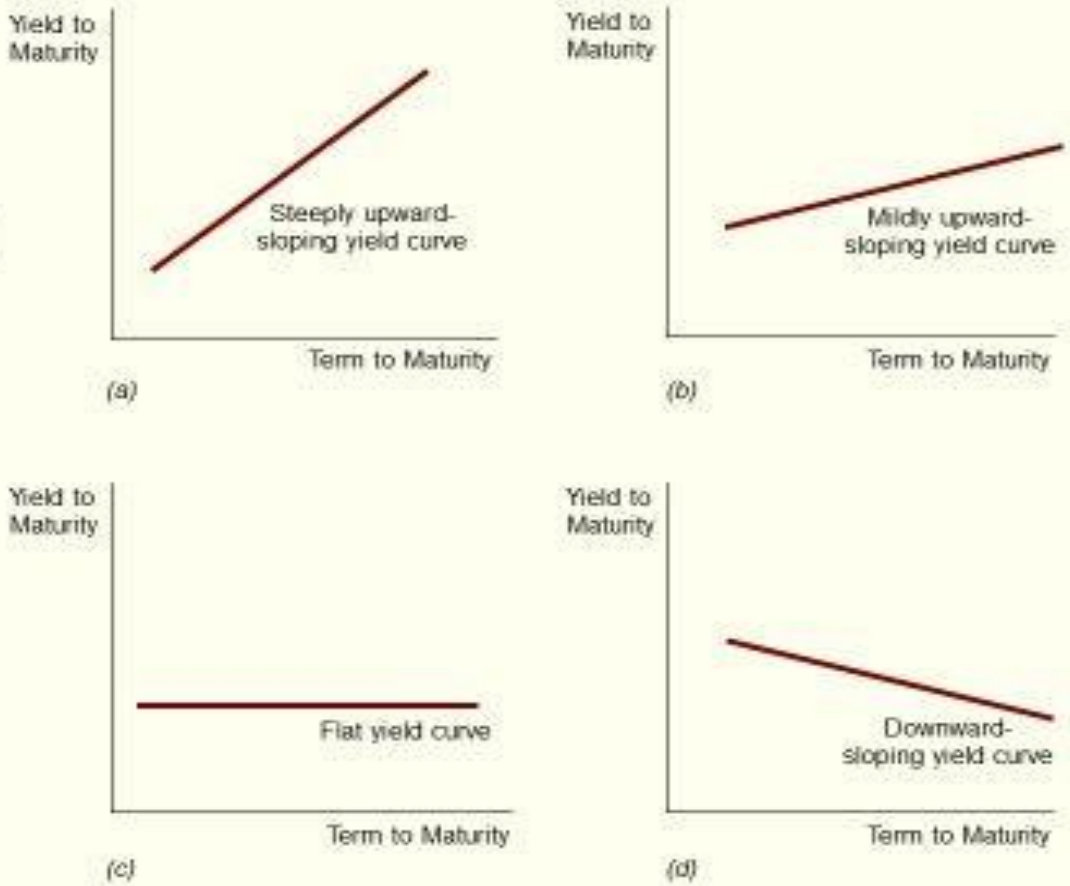
- ***Assumes that risk premium must necessarily rise with maturity because investors wish to liquidate their investments at the earliest and borrowers want to borrow long.***
- There is uncertainty about the one-year period return from a bond whose maturity is greater than one period. Uncertainty regarding the one-period return increases with the maturity of the bond.
- Since investors are risk-averse, they require an inducement to hold long-term bonds. They will ask for a long-term rate which is higher than the average of expected future rates.
- That means, forward rates should incorporate interest rate expectations + risk premium
- The yield curve will have a definite upward bias because investors show a preference for short-term securities.

The Relationship Between the Liquidity Premium (Preferred Habitat) and Expectations Theory



Using Yield Curve to Explain Liquidity Premium & Preferred Habitat Theories

Yield Curves and the Market's Expectations of Future Short-Term Interest Rates According to the Liquidity Premium (Preferred Habitat) Theory



Theories of Term Structure of Interest Rates

3. Market Segmentation Theory:

- An extreme format of the preferred habitat theory.
- It states that investors as well as borrowers are unwilling to shift from their preferred maturity range.
- Hence, the shape of the yield curve is determined entirely by the supply and demand forces within each maturity range**.**
- The yield curve can be just about any shape, depending on the supply and demand conditions.

Summary

Theory	Key Idea	How Yield Curve is Determined	Investor Behaviour / Assumptions
Unbiased Expectations Theory	Yield curve reflects market expectations of future short-term interest rates.	Long-term rate = average of expected future short-term rates.	Investors view long-term and successive short-term investments as perfect substitutes.
Liquidity Premium Theory	Long-term rates include a liquidity premium for higher risk and uncertainty.	Long-term rate = geometric average of current & expected short-term rates + liquidity premium that increases with maturity.	Investors accept long-term bonds only if compensated for additional market & liquidity risk.
Market Segmentation Theory	Investors prefer specific maturity ranges (“habitats”).	Interest rates determined separately in each maturity segment based on supply & demand.	Investors do not treat different maturities as substitutes; they invest according to their preferred horizons.

Applications of Yield Curve in Financial Management

The term structure of interest rates helps managers in

- valuation,
- financing decisions,
- cost of capital estimation,
- interest-rate risk management,
- portfolio strategy, and
- asset–liability matching.

Applications of Yield Curve in Financial Management

Area of Financial Management	Application of Term Structure	How It Helps in Decision-Making
Capital Budgeting	Used to discount long-term project cash flows using maturity-matched rates	Ensures more accurate NPV/IRR calculations and better project selection
Financing Decisions	Determines cost of issuing short-term vs. long-term debt	Helps firms choose optimal debt maturity structure and minimize financing cost
Capital Structure Planning	Identifies cheaper debt maturities based on the yield curve	Helps balance between short-term and long-term leverage
Bond Pricing & Valuation	Zero-coupon yield curve used to price bonds and fixed-income securities	Improves valuation accuracy and helps identify mispriced securities
Interest Rate Risk Management	Yield curve movements signal risk of refinancing and repricing	Guides duration matching, hedging, and swap strategies
Asset–Liability Management (ALM)	Banks, NBFCs, and insurers match asset and liability durations via term structure	Reduces mismatches that cause liquidity and interest-rate risk

Applications of Yield Curve in Financial Management

Area of Financial Management	Application of Term Structure	How It Helps in Decision-Making
Working Capital & Short-term Financing	Short-term rate segment helps decide on commercial paper, lines of credit, etc.	Minimizes short-term borrowing costs
Investment Portfolio Strategy	Guides decisions under expectations, liquidity preference, and market segmentation theories	Helps choose between long, medium, or short-term bonds
Monetary Policy Interpretation	Yield curve shape indicates future interest rates, inflation, and growth expectations	Aids in macro forecasting and strategic planning
Risk Pricing & Credit Spreads	Corporate yield curves compared with government yield curves	Helps estimate risk premiums and adjust cost of capital
Valuation of Derivatives	Term structure used to price interest-rate derivatives like swaps, caps, floors	Ensures accurate hedging and pricing for structured products
Foreign Borrowing Decisions	Comparing global yield curves to assess overseas financing options	Helps choose cheapest source of international funding

THANK YOU!!!